Chapter 3, Section 8

James Lee

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1. Let $f : X \to Y$ be a continuous map of topological spaces. Let $\mathcal{F}$ be a sheaf of Abelian groups on X, and assume that $R^i f_*(\mathcal{F}) = 0$ for all $i > 0$. Show that there are natural isomorphisms, for each $i \geqslant 0$,

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F}).$$

Proof. Let $0 \to \mathcal{F} \to \mathcal{I}^\bullet$ be an injective resolution of $\mathcal{F}$. By hypothesis, $0 \to f_*\mathcal{F} \to f_*\mathcal{I}^\bullet$ is also exact. The pushforward of an injective object in $\mathfrak{Ab}(X)$ is also injective in $\mathfrak{Ab}(Y)$ because f_* is a right adjoint functor, so $0 \to f_*\mathcal{F} \to f_*\mathcal{I}^\bullet$ is an injective resolution of $f_*\mathcal{F}$ in $\mathfrak{Ab}(Y)$. Suppose $\mathcal{G} \to \mathcal{G}'$ is a monomorphism of sheaves on Y, and $\mathcal{G} \to f_*\mathcal{I}$ is any morphism of sheaves. By adjunction, there is a morphism $f^{-1}\mathcal{G} \to \mathcal{I}$, and because f^{-1} is an exact functor, $f^{-1}\mathcal{G} \to f^{-1}\mathcal{G}'$ is also a monomorphism (note that in general, f^* is only right exact). By injectivity of $\mathcal{I}$, there is an extension of $f^{-1}\mathcal{G} \to \mathcal{I}$ making the following diagram commute

$$\begin{tikzcd} & \mathcal{I} \\ f^{-1}\mathcal{G} \arrow[ur] \arrow[r] & f^{-1}\mathcal{G}' \arrow[u, dashed] \end{tikzcd}.$$

Hence, by adjunction again, there is an extension of $\mathcal{G} \to f_*\mathcal{I}$ making the following diagram commute

$$\begin{tikzcd} & f_*\mathcal{I} \\ \mathcal{G} \arrow[ur] \arrow[r] & \mathcal{G}' \arrow[u, dashed] \end{tikzcd}.$$

Since $\mathcal{F}$ and $f_*\mathcal{F}$ have same global sections for any $\mathcal{F}$, there are natural identificiations

$$\begin{tikzcd} 0 \arrow[r] & \Gamma(X, \mathcal{F}) \arrow[r] \arrow[d, equal] & \Gamma(X, f_*\mathcal{I}^\bullet) \arrow[d, equal] \\ 0 \arrow[r] & \Gamma(Y, f_*\mathcal{F}) \arrow[r] & \Gamma(Y, f_*\mathcal{I}^\bullet) \end{tikzcd}.$$

Taking cohomology gives the desired natural isomorphisms for all $i \geqslant 0$. ☐

2. Let $f : X \to Y$ be an affine morphism of schemes (II, Ex. 5.17) with X Noetherian, and let $\mathcal{F}$ be a quasi-coherent sheaf on X. Show that the hypotheses of (Ex. 8.1) are satisfies, and hence that $H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F})$ for each $i \geqslant 0$.

Proof. Let V be an open affine subset of Y. By (8.2) and (8.5), we have

$$R^i f_*(\mathcal{F})\Big|_V \cong \widetilde{H^i(f^{-1}V, \mathcal{F}|_{f^{-1}(V)})},$$

which is zero because $f^{-1}(V)$ is affine (3.7). Hence, $R^i f_*(\mathcal{F}) = 0$, and the result follows by the previous exercise. ☐

3. Let $f : X \to Y$ be a morphism of ringed spaces, let $\mathcal{F}$ be an $\mathcal{O}_X$-module, and let $\mathcal{E}$ be a locally free $\mathcal{O}_Y$-module of finite rank. Prove the *projection formula* (cf. (II, Ex. 5.1))

$$R^i f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong R^i f_*(\mathcal{F}) \otimes \mathcal{E}.$$

Proof. The pullback of a locally free sheaf of finite rank is also locally free of finite rank. Localization is an exact monoidal functor from the category of $\mathcal{O}_Y$-modules to the category of Abelian groups, so by checking exactness at stalks, tensoring by $\mathcal{E}$ is an exact functor, and it also has a left adjoint, where for any $\mathcal{O}_Y$-modules $\mathcal{M}$ and $\mathcal{N}$ (II, Ex. 5.1)

$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M}\otimes\mathcal{E}^\vee,\mathcal{N}) &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M},\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{E}^\vee,\mathcal{N}))\\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M},\mathcal{N}\otimes(\mathcal{E}^\vee)^\vee)\\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M},\mathcal{N}\otimes\mathcal{E}).\end{aligned}$$

Hence, the result follows from the $i=0$ case, (1.3A), and Lemma 1.

It remains to show

$$f_*(\mathcal{F}\otimes f^*\mathcal{E})\cong f_*(\mathcal{F})\otimes\mathcal{E}.$$

We first show there is a natural map

$$\phi: f_*(\mathcal{F})\otimes\mathcal{E}\to f_*(\mathcal{F}\otimes f^*\mathcal{E}).$$

Let $\mathcal{P}$ be the presheaf on Y defined for open subsets $V\subseteq Y$ by

$$V\mapsto f_*(\mathcal{F})(V)\otimes\mathcal{E}(V)$$

so that $\mathcal{P}^{sh}\cong f_*(\mathcal{F})\otimes\mathcal{E}$, and let $\mathcal{Q}$ be the presheaf on X defined for open subsets $U\subseteq X$ by

$$U\mapsto\mathcal{F}(U)\otimes f^*\mathcal{E}(U)$$

so that $\mathcal{Q}^{sh}\cong\mathcal{F}\otimes f^*\mathcal{E}$. Sheafification and pushforward do not commute; nonetheless there is a natural morphism $(f_*\mathcal{Q})^{sh}\to f_*(\mathcal{Q}^{sh})$. In particular, consider the natural morphism $\mathcal{Q}\to\mathcal{Q}^{sh}$. Taking the pushforward gives $f_*\mathcal{Q}\to f_*(\mathcal{Q}^{sh})$. Because $f_*(\mathcal{Q}^{sh})$ is already a sheaf, by the universal property of sheafification (II, 1.2), there is unique map $\tau:(f_*\mathcal{Q})^{sh}\to f_*(\mathcal{Q}^{sh})$ fitting into the commutative diagram

$$\begin{tikzcd} & (f_*\mathcal{Q})^{sh} \arrow[dr, dashed, "\tau"] & \\ f_*\mathcal{Q} \arrow[ur] \arrow[rr] & & f_*(\mathcal{Q}^{sh}) \end{tikzcd}.$$

Thus, we define a morphism of presheaves $\psi:\mathcal{P}\to f_*\mathcal{Q}$, which induces a morphism of sheaves $\psi^{sh}:\mathcal{P}^{sh}\to(f_*\mathcal{Q})^{sh}$, and composing with τ gives our desired morphism

$$\phi=\tau\circ\psi^{sh}:\mathcal{P}^{sh}\to(f_*\mathcal{Q})^{sh}\to f_*(\mathcal{Q}^{sh}).$$

The pushforward for the presheaf $\mathcal{Q}$ distributes over the tensor product, that is

$$\Gamma(V,f_*\mathcal{Q})=f_*(\mathcal{F})(V)\otimes f_*f^*(\mathcal{E})(V).$$

There is a natural morphism of sheaves $\iota:\mathcal{E}\to f_*f^*(\mathcal{E})$, so take $\psi=\mathrm{id}_{f_*\mathcal{F}}\otimes\iota$ to be our morphism of presheaves

$$\begin{tikzcd} \mathcal{P}^{sh} \arrow[r, "\psi^{sh}"] \arrow[rr, bend left, "\phi"] & (f_*\mathcal{Q})^{sh} \arrow[r, "\tau"] & f_*(\mathcal{Q}^{sh}) \\ \mathcal{P} \arrow[u] \arrow[r, "\psi"'] & f_*\mathcal{Q} \arrow[u] \arrow[ur] & \end{tikzcd}.$$

Now that we defined a morphism, we remark that the question is local on Y. In particular, let y be a point in Y, and let V be an open subset of Y such that $\mathcal{E}$ is free of finite rank. Then $f^*\mathcal{E}$ is free of finite rank on $U=f^{-1}(V)$. Consider the commutative diagram

$$\begin{tikzcd} U \arrow[r, "\bar{f}"] \arrow[d, "i"'] & V \arrow[d, "j"] \\ X \arrow[r, "f"] & Y \end{tikzcd}.$$

Because i and j are open immersions, there is an isomorphism of functors

$$j^*f_* \cong \bar{f}_*i^*, \quad i^*f^* \cong \bar{f}^*j^*.$$

Pullback distributes over tensor products, so we have

$$\begin{aligned}
j^*f_*(\mathcal{F} \otimes f^*\mathcal{E}) &\cong \bar{f}_*i^*(\mathcal{F} \otimes f^*\mathcal{E}) \\
&\cong \bar{f}_*(i^*\mathcal{F} \otimes i^*f^*\mathcal{E}) \\
&\cong \bar{f}_*(i^*\mathcal{F} \otimes \bar{f}^*j^*\mathcal{E}), \\
j^*(f_*(\mathcal{F}) \otimes \mathcal{E}) &\cong j^*f_*(\mathcal{F}) \otimes j^*\mathcal{E} \\
&\cong \bar{f}_*(i^*\mathcal{F}) \otimes j^*\mathcal{E}.
\end{aligned}$$

Open immersions preserve stalks, so we can assume $\mathcal{E}$ to be a free $\mathcal{O}_X$-module of finite rank n. The pullback of a free sheaf of finite rank is also free of finite rank, so we have the isomorphisms

$$\begin{aligned}
f_*(\mathcal{F} \otimes f^*\mathcal{E}) &\cong f_*(\mathcal{F} \otimes \mathcal{O}_X^{\oplus n}) \\
&\cong f_*(\mathcal{F}^{\oplus n}) \\
&\cong f_*(\mathcal{F})^{\oplus n} \\
&\cong f_*(\mathcal{F}) \otimes \mathcal{O}_Y^{\oplus n} \\
&\cong f_*(\mathcal{F}) \otimes \mathcal{E}.
\end{aligned}$$

The third isomorphism holds because f_* is right adjoint, and right adjoint functors preserve limits, and a direct sum of finite number of terms can be realized as a limit, so the pushforward and finite direct sums commute. □

Lemma 1. *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be Abelian categories, and assume $\mathcal{A}$ and $\mathcal{B}$ have enough injectives. Let $F : \mathcal{A} \to \mathcal{B}$ and $G : \mathcal{B} \to \mathcal{C}$ be functors such that F has a left adjoint (or more generally preserves injectives), and either*

(a) F *is left exact, and* G *is exact; or*

(b) F *is exact, and* G *is left exact.*

In each case, we have the following natural isomorphisms

(a) $R^i(G \circ F) \cong G \circ R^i(F)$*;*

(b) $R^i(G \circ F) \cong R^i(G) \circ F$.

Proof. The proof for (b) is similar to the proof of Exercise 1, so we prove (a). Let A be an object in $\mathcal{A}$ and choose an injective resolution

$$0 \to A \to I^\bullet,$$

where one has

$$R^i F(A) = h^i(FI^\bullet).$$

Applying $G \circ F$ gives a complex in $\mathcal{C}$

$$0 \to (G \circ F)A \to (G \circ F)I^\bullet,$$

and we want to show

$$h^i((G \circ F)I^\bullet) \cong G \circ h^i(FI^\bullet).$$

But this follows from exact functors commuting with cohomology.

□